No four coplanar in the cube: a witness for $a(7) \geq 18$ and the state of the upper bound

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Abstract

Let $a(n)$ be the largest number of points of $\{0, \dots, n-1\}^3$ no four of which are coplanar, OEIS A280537. The entry records $5, 8, 10, 13, 16, 18, 20$ for $n = 2, \dots, 8$ and states that the terms up to $a(6)$ were found by exhaustive search while $a(7)$ and $a(8)$ rest on numerical evidence. We reproduce $a(2), \dots, a(5)$ by three independent methods — $a(5) = 13$ with an unsatisfiability certificate verified by an external checker — exhibit witnesses for $a(6) \geq 16$ and $a(7) \geq 18$, which the entry did not carry, and report the state of the remaining question, the impossibility of 19 points at $n = 7$.

We also record how we nearly published the opposite. Having closed 124 of 128 regions of a case split at $M = 18$ without finding a configuration, we leaned towards $a(7) = 17$ and towards announcing that the published value is wrong. The four unclosed regions were precisely the ones the symmetry pruning drives solutions into, and the configuration was in them. It was found by an independent search and verified here three ways. Nothing in our computations was incorrect; what was incorrect was reading the fraction of closed regions as an approximation to a conclusion.

1 Reformulation

Four points are coplanar exactly when they lie in a common plane, and a plane carrying at most three lattice points forbids nothing. Hence

$$S \subseteq [n]^3 \text{ is admissible} \iff |S \cap P| \leq 3 \text{ for every plane } P \text{ with} \geq 4 \text{ lattice points,}$$

a propositional formula on n^3 variables. The layer $z = k$ is itself such a plane, so $a(n) \leq 3n$; the published values satisfy $5 \leq 6$, $8 \leq 9$, $10 \leq 12$, $13 \leq 15$, $16 \leq 18$, $18 \leq 21$, $20 \leq 24$.

2 What is established here

n	value	how
2	$a = 5$	three independent methods
3	$a = 8$	three independent methods; solution count matches the published one
4	$a = 10$	three methods; **232 classes**, matching the published count
5	$a = 13$	three methods; unsatisfiability at 14 carries a certificate verified by `drat-trim`
6	$a \geq 16$	four independent witnesses, no two equivalent under the cube group
7	$a \geq 18$	witness, verified three ways here and four ways independently

The strongest of these checks is not that the maxima agree but that the *counts* do. Ed Pegg reported a unique 8-point configuration at $n = 3$ and 232 configurations of 10 points at $n = 4$; our enumerator gives 16 and 10960 labelled configurations, collapsing to exactly 1 and 232 classes under the 48 symmetries of the cube, with stabiliser split $225 \cdot 48 + 6 \cdot 24 + 1 \cdot 16 = 10960$. Matching a maximum can happen by accident; matching a count cannot.

3 What is *not* established: there are no upper bounds here

This deserves its own paragraph, because a reader who has seen the companion note on the no-three-in-line problem — where the upper bounds are the substance — will otherwise supply one by habit.

For the present problem we have **no completed upper bound above** $n = 5$. At $n = 5$ the impossibility of 14 points is established and carries a certificate. At $n = 6$ we have four witnesses for 16 and a case split at $M = 17$ that is not finished; the published $a(6) = 16$ rests on someone else's exhaustive search, not on ours. At $n = 7$ we have six witnesses for 18 and a case split at $M = 19$ in progress. Every entry of the table above that is written $a \geq$ is a lower bound and nothing more.

We say this explicitly because the tempting summary — "we settled $a(7)$" — would be false in the half that is hard. Exhibiting a configuration is cheap and self-verifying; showing that none is larger is neither.

4 The near-miss, in full

We record this because the mechanism is more transferable than the result.

The decisive question was split into regions by the contents of the first columns; a column cannot hold four points, since four points of a column are collinear and hence coplanar, so subsets of size at most three exhaust it. At $M = 18$ we closed 124 of 128 regions with no configuration found, and reported the fraction as though it were progress towards a conclusion.

It was not. The pruning we use keeps, from each orbit of the cube group, the lexicographically least member — the one with zeros as early as possible. Solutions are therefore driven systematically into the least constrained regions, which are exactly the ones that close last. The unclosed regions were not a remainder; they were the whole question. We had written this out for each other two hours earlier and still read the fraction the other way.

An independent search then produced three 18-point configurations. We verified one here by three separate computations: all $\binom{18}{4} = 3060$ quadruples with the 3×3 determinant of differences, all 3060 with the 4×4 determinant of homogeneous coordinates, and membership in our list of 346743 rich planes at $n = 7$, itself verified complete against 6650464 non-collinear triples. All three give zero.

What saved us was not caution. It was that neither of us ever wrote "does not exist" where we meant "have not found".

5 The remaining question

Whether 19 points fit at $n = 7$ is open at the time of writing and is being decided by exhaustive case split. We will not report a fraction of it.

AI/tool-use disclosure

The searches, the programs and the text were produced by autonomous AI agents (Anthropic Claude) under the direction of the author of record, who posed the question, chose what to compute, decided what to claim and takes responsibility for the content. Verification protocol, programs, journals and witnesses: `https://github.com/iwasborninbali/saturation`.